Chunghyun Park

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Postdoctoral researcher, POSTECH Institute of AI
Citizenship: Republic of Korea

Research interests

3D visual perception and scene understanding, spanning geometric deep learning, open-vocabulary 3D segmentation, and robotic manipulation.

Education

2022 – 2026 **Pohang University of Science and Technology (POSTECH)**
Ph.D. in Artificial Intelligence
Thesis: Learning 3D Visual Perception for Geometric and Semantic Generalization
Advisor: Prof. Minsu Cho

2020 – 2022 **Pohang University of Science and Technology (POSTECH)**
M.S. in Artificial Intelligence
Thesis: Fast Point Transformer for Large-scale 3D Scene Understanding
Advisor: Prof. Jaesik Park

2014 – 2019 **Pohang University of Science and Technology (POSTECH)**
B.S. in Mechanical Engineering

Experience

Mar. 2026 – Present **POSTECH Institute of AI** (Postdoctoral researcher) – Pohang, Republic of Korea
Host: Prof. Minsu Cho
Enhancing 3D understanding of VLA models for robotic manipulation.

Dec. 2023 – Feb. 2025 **NVIDIA** (Research intern) – Taipei, Taiwan (Remote)
Hosts: Jaesung Choe, Yu-Chiang Frank Wang
Researched open-vocabulary 3D segmentation with the NVIDIA LPR team (Junha Lee, Chris Choy, Jan Kautz), publishing Mosaic3D at CVPR 2025.

Publications

** indicates equal contribution.*

2026 **SpaCeFormer: Fast Proposal-Free Open-Vocabulary 3D Instance Segmentation**
Chris Choy, Junha Lee, Chunghyun Park, Minsu Cho, Jan Kautz
arXiv

2026 **Affostruction: 3D Affordance Grounding with Generative Reconstruction**
Chunghyun Park, Seunghyeon Lee, Minsu Cho
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)

2025 **Affogato: Learning Open-Vocabulary Affordance Grounding with Automatic Data Generation at Scale**
Junha Lee*, Eunha Park*, Chunghyun Park, Dahyun Kang, Minsu Cho
arXiv

2025 **Combinative Matching for Geometric Shape Assembly**
Nahyuk Lee*, Juhong Min*, Junhong Lee, Chunghyun Park, Minsu Cho
IEEE/CVF International Conference on Computer Vision (ICCV) Highlight

2025 **Mosaic3D: Foundation Dataset and Model for Open-Vocabulary 3D Segmentation**
Junha Lee*, Chunghyun Park*, Jaesung Choe, Yu-Chiang Frank Wang, Jan Kautz, Minsu Cho, Chris Choy
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)

2024 **Learning SO(3)-Invariant Semantic Correspondence via Local Shape Transform**
Chunghyun Park*, Seungwook Kim*, Jaesik Park, Minsu Cho
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)

2023 **Stable and Consistent Prediction of 3D Characteristic Orientation via Invariant Residual Learning**
Seungwook Kim*, Chunghyun Park*, Yoonwoo Jeong, Jaesik Park, Minsu Cho
International Conference on Machine Learning (ICML)

2022 **SeLCA: Self-Supervised Learning of Canonical Axis**
Seungwook Kim*, Yoonwoo Jeong*, Chunghyun Park*, Jaesik Park, Minsu Cho
NeurIPS Workshop on Symmetry and Geometry in Neural Representations

2022 **PointMixer: MLP-Mixer for Point Cloud Understanding**
Jaesung Choe*, Chunghyun Park*, Francois Rameau, Jaesik Park, In So Kweon
European Conference on Computer Vision (ECCV)

2022 **Fast Point Transformer**
Chunghyun Park, Yoonwoo Jeong, Minsu Cho, Jaesik Park
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)

2021 **Improved Classification and Localization Approach to Small Bowel Capsule Endoscopy using Convolutional Neural Network**
Yunseob Hwang*, Han Hee Lee*, Chunghyun Park, Bayu Adhi Tama, Jin Su Kim, Dae Young Cheung, Woo Chul Chung, Young-Seok Cho, Kang-Moon Lee, Myung-Gyu Choi, Seungchul Lee, Bo-In Lee
Digestive Endoscopy

Patents

2025 **Hardware Apparatus and Method for Predicting Structural Characteristic Orientation of Point Cloud of Structural Object**
Seungwook Kim*, Chunghyun Park*, Yoonwoo Jeong, Jaesik Park, Minsu Cho
U.S. Patent Application No. US 2025/0191218 A1

Awards & honors

2026 CVPR Doctoral Consortium
Awarded to selected Ph.D. students in computer vision worldwide

2026 Bronze Prize, Best Paper Award (IPIU)
Top papers selected from the annual Korean workshop

2025 POSTECHIAN Fellowship – Innovation (POSTECH)
For outstanding graduate students of the semester (₩6,000,000)

2024 Outstanding Reviewer (CVPR)
Top 2% of 9,872 reviewers

2023 Grand Prize, BK21 Best Paper Award (POSTECH GSAI)
Best paper of the year at POSTECH GSAI

2022 Qualcomm Innovation Fellowship (Qualcomm Korea)
Top AI papers by graduate students in South Korea (₩4,000,000)

2022 Silver Prize, Samsung HumanTech Paper Award (Samsung Electronics)
Top papers across 10 fields nationwide (₩7,000,000)

Talks

Apr. 2026 Generalizable 3D Perception for Physical AI
Invited talk, Chung-Ang University
Hosts: Prof. Jongmin Lee, Prof. Sooyoung Lee

Mar. 2024 Equivariant Networks for 3D Vision
Guest lecture, CSED703J Symmetry and Equivariant Learning

Nov. 2023 Stable and Consistent Prediction of 3D Characteristic Orientation
Spotlight presentation, Korea AI Summit

Professional services

Journal Reviewer
IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)
International Journal of Computer Vision (IJCV)

Conference Reviewer

IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)
IEEE/CVF International Conference on Computer Vision (ICCV)
European Conference on Computer Vision (ECCV)
Conference on Neural Information Processing Systems (NeurIPS)
International Conference on Learning Representations (ICLR)